

An Exploration of Sentiment Analysis Methods to Study Student Perspectives on Artificial Intelligence

This summer, I worked with Professor Fourquet to learn how to conduct sentiment analysis on text using the programming language R. Sentiment analysis is the process of computationally deriving emotional tone from a text, such as determining if a text has a more positive or negative tone. The data that we used to aid our learning was collected by COSC 101 students at the end of the spring 2025 semester. Students submitted short videos discussing their thoughts on AI and whether or not they think that AI could be intelligent. Professor Fourquet turned these videos into text files using a text transcription software. Then, I cleaned the text files by correcting any incorrect word transcriptions and organizing the text in a clear manner. Throughout the summer I used these text files to conduct a pilot experiment of what sentiment analysis can tell us about student thoughts on AI. The goal of this experiment is to learn sentiment analysis techniques that will aid Professor Fourquet's future plans to study how student sentiments towards AI change after taking an introductory computer science course (COSC 101). She will do this by asking students to submit a video on their thoughts about AI at the beginning and the end of the semester, and then analyzing similarities and differences in student sentiments.

To begin our research, I learned key functions of R such as fundamental libraries to download and basic data visualization methods. I also wrote code using prebuilt datasets to familiarize myself with R. Next, I researched how text files can be interpreted into data frames. This research helped me code a program that reads a text file and outputs a visualization of the most frequent words in the text (excluding numbers, words less than four letters, and stop words such as "and," "the," and "of"). Then I learned about the Bing sentiment lexicon, which is a prebuilt list of words labeled as positive or negative. Using this lexicon, I created the following data visualizations from a single student text file: a bar chart of the number of positive vs. negative words, a horizontal bar chart of which words contribute the most to the positive and negative word counts, and a visualization of how the positive and negative sentiment changes over time. Next, I created a large text file compiled of twelve student text transcriptions. I used this comprehensive text file to create two wordclouds, which are data visualizations of text where the size of each word indicates its frequency or importance. These word clouds represent the general student sentiments towards AI. To conclude our research, I created a Quarto document in R that thoroughly explains my research on sentiment analysis and the steps to code each data visualization that I made.

